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To cite this article: Kristin Østlie, Ingrid Marie Lesjø, Rosemary Joy Franklin, Beate Garfelt, Ola Hunsbeth Skjeldal & Per Magnus (2012) Prosthesis rejection in acquired major upper-limb amputees: a population-based survey, *Disability and Rehabilitation: Assistive Technology*, 7:4, 294-303, DOI: [10.3109/17483107.2011.635405](https://doi.org/10.3109/17483107.2011.635405)

To link to this article: <https://doi.org/10.3109/17483107.2011.635405>



Published online: 23 Nov 2011.



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RESEARCH PAPER

Prosthesis rejection in acquired major upper-limb amputees: a population-based survey

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Purpose: To estimate the rates of primary and secondary prosthesis rejection in acquired major upper-limb amputees (ULAs), to describe the most frequently reported reasons for rejection and to estimate the influence of background factors on the risk of rejection. **Method:** Cross-sectional study analysing population-based questionnaire data ($n = 224$). Effects were analysed by logistic regression analyses and Cox regression analyses. **Results:** Primary prosthesis rejection was found in 4.5% whereas 13.4% had discontinued prosthesis use. The main reasons reported for primary non-wear were a perceived lack of need and discrepancies between perceived need and the prostheses available. The main reasons reported for secondary prosthesis rejection were dissatisfaction with prosthetic comfort, function and control. Primary prosthesis rejection was more likely in ULAs amputated at high age and in ULAs with proximal amputations. Secondary prosthesis rejection was more likely in proximal ULAs and in women. **Conclusions:** Clinicians should be aware of the increased risk of rejection in proximal ULAs, elderly ULAs and in women. Emphasising individual needs will probably facilitate successful prosthetic fitting. Improved prosthesis quality and individualised prosthetic training may increase long-term prosthesis use. Further studies of the effect of prosthetic training and of the reasons for rejection of different prosthetic types are suggested.

Keywords: Amputees, multivariate analysis, prosthesis use, rehabilitation, rejection, upper limb

Introduction

Arms and hands are used for most activities of daily living and are also important for communication and affection [1,2]. Major amputation (through or proximal to the wrist) causes a total loss of hand and wrist function on the affected side [3], resulting in significant disability [2,4]. The

Implications for Rehabilitation

- Most acquired major upper-limb amputees (ULAs) are fitted with prostheses after the amputation.
- This population-based study shows that proximal ULAs, elderly ULAs and women have an increased risk of prosthesis rejection.
- Emphasising individual needs may facilitate successful prosthetic fitting.
- Improved prosthesis quality and individualised prosthetic training may increase long-term prosthesis use.

importance of optimal functional rehabilitation in these patients is emphasised by upper-limb amputees (ULAs) generally being young, otherwise healthy and having many years of productive employment and active contribution to the society ahead [5].

Arm prostheses are an important aid to function for ULAs, and in our experience, most major ULAs are fitted with prostheses after the amputation [6]. The percentage of long-term users reported however varies considerably [7]; prosthetic usage rates vary from 39% to 81% in Northern hemisphere studies [8]. This wide range of variance may be due to the heterogeneous samples involved and methodological differences between studies and/or characteristics of the clinical programmes or the prostheses involved [7]. There is a lack of standardised outcome measurement tools regarding prosthetic use in ULAs [7], [9–12], and the comparison of the degree of use between studies is difficult as the categories of wearing time in published reports are varied or poorly defined [5].

The variety of activities performed by the upper limb and the dexterity and sensory capacity of the hand pose major challenges to functional restoration [5], and the field of upper-limb prosthetics is a constantly changing arena as researchers

and prosthetists strive to bridge the gap between prosthetic reality and upper-limb physiology [13]. For prosthesis-rejecting ULAs, the prosthesis does not offer sufficient functional advantage to compensate for any inconvenience or discomfort involved in its use [13,14]. An overview of the literature regarding the factors associated with prosthesis use and abandonment was given in a review by Biddiss and Chau in 2007 [9]. While the influence of some factors (i.e. lifestyle, level of limb loss) is strongly supported in the literature, the impact of others (i.e. age of fitting, efficacy of training protocols) remains controversial and the interplay between these factors and their relative importance in prosthesis rejection is not well understood [9]. Furthermore, it is likely that the reasons for primary rejection (never worn prostheses) are somewhat different from the reasons for secondary rejection (discontinued wear). An enhanced understanding of the factors affecting prosthesis use and abandonment is needed to optimise prescription practices, guide design efforts and satiate demand for evidence-based measures of intervention [7,9].

In the existing literature, there is a lack of population-based studies regarding prosthesis use and rejection. The majority of earlier studies have focused on an isolated population contacted through a single rehabilitation centre, limiting the generality of the results [7]. In the existing literature, there is also a lack of comprehensive, descriptive documentation and of multivariate analyses which would facilitate the uncovering of main and interaction effects on rejection rates, and there is a paucity of studies conducting formal statistical analyses necessary to ensure that observations are not merely a manifestation of random effects [7].

Aims of the study

The aims of this study were to use population-based questionnaire data (1) to estimate the rates of primary and secondary prosthesis rejection in acquired major ULAs, (2) to describe the most frequently reported reasons for rejection and (3) to use multivariate analyses to estimate the influence of background factors on the risk of rejection.

Methods

Design and subjects

We performed a cross-sectional study. Eligible amputees from the known population of adult acquired major ULAs in Norway were invited to participate in the study. This amputee population was identified through a combined search of the databases of the two companies in Norway that make upper-limb prostheses: Norwegian Technical Orthopaedics AS (NTO AS) and Sophies Minde Orthopaedics AS (SMO AS) and of the medical records of three major Norwegian hospitals: Oslo University Hospital Rikshospitalet, St. Olavs Hospital (Trondheim) and Haukeland University Hospital (Bergen). The inclusion process is described in detail in a previous paper [6].

The inclusion period was from 1 October 2006 until 30 May 2008. Inclusion criteria were: acquired upper-limb loss, major amputation (through or proximal to the radio carpal joint),

adult (≥ 18 years of age at 1 October 2006), resident in Norway at survey and mastering spoken and written Norwegian. An exclusion criterion was inability to give the information required (i.e. dementia, aphasia).

Data collection

Data from all eligible amputees were collected by postal questionnaires returned by pre-paid mail. Prior to data collection, our study was approved by the Norwegian Regional Ethics Committee. For the search for amputees in hospital medical records, we gained exemption from the duty of confidentiality by The Norwegian Directorate for Health and Social Affairs.

Variables

We collected data on demographic factors, amputation-related factors and prosthesis-related factors (background factors). Demographic factors were: gender, age at survey, ethnicity (native country), place of residence at survey (region), occupational status at amputation and gross yearly income during the last year before amputation. Amputation-related factors were: age at amputation, date of amputation, unilateral/bilateral amputation, level of amputation, side of amputation, amputation of dominant hand and cause of amputation. Prosthesis-related factors were: date of first prosthetic fitting, date of final rejection, prosthetic training and prosthetic types worn at survey or before rejection. We also collected data on prosthesis use and rejection, first referral (by whom) and on self-reported reasons for prosthesis rejection.

Prosthetic training was specified as practical training in prosthesis use performed together with a professional. It was also specified that information on prosthetic use without practical training should not be considered as prosthetic training. The amputees were then asked 'Have you ever had prosthetic training?' (yes/no). The amputees were also asked whether they had had enough prosthetic training to meet their needs (yes/no).

Five different prosthetic types were clearly defined: myoelectric prostheses, body-powered prostheses, cosmetic prostheses, aesthetic prostheses and hybrid prostheses, and the amputees were asked to give information on the type(s) of prostheses worn at survey or before rejection. There was also a space for comments on this in the questionnaire.

Asking the amputees to specify the reason(s) for secondary prosthesis rejection, problems within five aspects of prosthesis use were defined as follows: (1) Appearance: Colour, design, size, degree of natural appearance, judgement of others. (2) Comfort: Weight, socket fit, sweating, comfort of the harness and the wires, donning and doffing of the prosthesis. (3) Function: The ability to grip, support and stabilise, wrist motion, speed for opening and closing of the prosthetic hand, noise. (4) Control: Control of opening and closing of the prosthetic hand, unintended movements, the ability to prevent objects from slipping out of the prosthetic hand, difficulty of prosthesis use. (5) Reliability: Durability of the harness, wires, gloves and batteries. How easily the prosthesis stains, easiness

of cleaning. Other possible reasons specified in the questionnaire were: lack of information on prostheses and prosthesis use, lack of prosthetic training and insufficient availability of prosthetic adjustment. For primary prosthesis rejection, possible reasons specified in the questionnaire were: medical causes (including amputation level and co-morbidities), failure of the prostheses available to meet needs, total lack of need, lack of information on prostheses and prosthesis use, prosthetic fitting not offered, low availability of prosthetic services, low availability of prosthetic training and cultural or religious causes. There was also a space for comments on both types of prosthesis rejection in the questionnaire.

Data analyses

Data were processed using the Statistical Package for the Social Sciences (SPSS) version 14.0 as a general rule. Crude odds ratios (ORs) for prosthesis rejection in unilateral vs bilateral amputees and for prosthesis rejection in child/youth/students vs those in paid employment were found by an exact method using the statistical package Stata.

95% confidence intervals (CIs) for were calculated with a binomic formula and for these CIs negative limits were rounded up to 0.0 and limits >100 were rounded down to 100.0. The precision of the calculated effect estimates are given as 95% CIs. We did not perform hypothesis testing for effect estimates as a general rule. However, for comparing questionnaire responders and non-responders we used hypothesis testing and the significance level was set at $\alpha=0.05$.

To assess the generality of our findings we compared the questionnaire responders ($n=224$) and non-responders ($n=166$) regarding the variables known for both groups: gender, age at survey, place of residence, ethnicity, unilateral/bilateral amputation, side of amputation and level of amputation (above vs below-elbow). We used χ^2 Chi-square tests for frequencies and one-way ANOVA analyses for continuous variables.

Based on our questionnaire data we calculated the time from the amputation until the first prosthetic fitting and the time from the first prosthetic fitting to rejection or to survey. Due to a high proportion of missing data especially regarding day and month of the first prosthetic fitting, imputation was performed before calculating the time from the amputation to the first prosthetic fitting. For 69 amputees for whom the year of amputation and the year of first fitting were known, the following imputation-scheme was used: Unknown day of month was set as 15th. Unknown month was set as June. If amputation was performed after June 15th and prostheses fitted in the same year in an unknown month, a maximum estimate of time to first fitting was given by the number of remaining months of the year. Further details on the imputation-scheme can be obtained from the authors on request.

The rejection rates and the reasons for prosthesis rejection specified by the rejecting amputees were reported.

The effects of background factors on prosthesis rejection were analysed separately for primary and secondary rejection, defined as follows: (1) Primary prosthesis rejection: The amputee has never worn prostheses. (2) Secondary prosthesis rejection: Rejection after some time of prosthesis

use (see Figure 1). The functional needs and prosthetic rehabilitation of unilateral and bilateral amputees are known to be very different [5,14], calling for separate analyses of effects on prosthesis rejection in these subgroups of ULAs. Due to the small sample-size of bilateral ULAs in our material ($n=10$), multivariate analyses were only performed in unilateral ULAs.

In unilateral amputees ($n=209$), relative risks (RRs) for primary prosthesis rejection were approximated by calculating ORs with logistic regression analyses. The dependent variable was 'Primary prosthesis rejection' with the categories 'use' ($n=199$) and 'rejection' ($n=10$). Five unilateral amputees waiting for their first prosthesis at survey were excluded from these analyses as they could not be allocated to any group. The demographic and amputation-related factors analysed are displayed in Table I. Crude OR estimates (one independent variable in the model at a time) were calculated for all the covariates. Calculating adjusted OR estimates (i.e. estimates adjusted for confounding effects of the other covariates), only the variables 'gender', 'ethnicity', 'age at amputation', 'level of amputation' and 'amputation of dominant hand' were included, since the inclusion of other variables did not lead to a valid model.

In unilateral amputees ($n=199$), RRs for secondary prosthesis rejection were estimated using Cox regression. 'Time' was defined as time from first prosthetic fitting until final rejection for rejecters ($n=27$) and as time from first prosthetic fitting to time of survey (censored) for those still wearing prostheses ($n=172$). The event defined was 'Secondary prosthesis rejection'. The independent variables analysed were as for primary prosthesis rejection, plus the prosthesis-specific variables: 'time from amputation until first prosthetic fitting', 'prosthetic training' (yes/no), 'prosthetic type' and 'prosthetic training enough to meet needs' (yes/no). First, Cox regression analyses were conducted calculating crude and adjusted effect estimates for all covariates except for 'prosthetic type' (Table II). The multivariate analysis was conducted both including and excluding the variable 'prosthetic training enough to meet needs' due to a high rate of missing for this variable ($n=44$). The variable is of clinical importance and as the other effect

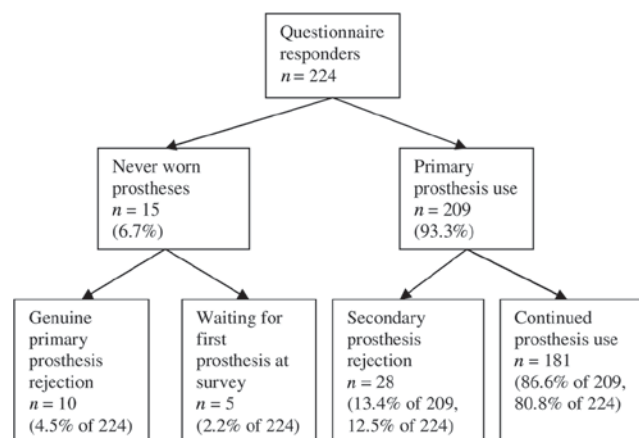


Figure 1. Prosthesis use, primary and secondary prosthesis rejection among the questionnaire responders.

Table I. Primary prosthesis rejection: effects of demographic and amputation-related factors in unilateral amputees ($n = 209^a$).

Independent variable	Primary prosthesis rejection			Crude effect estimates ^b		Adjusted effect estimates ^c	
	Use (n) ^d	Rejection (n) ^e	% rejection	cOR	95% CI for cOR	aOR	95% CI for aOR
Gender							
Men	167	7	4.0	1.00	Ref	Ref	Ref
Women	32	3	8.6	2.24	0.55–9.11	1.06	0.18–5.96
Ethnicity (native country)							
Norway	182	9	4.7	1.00	Ref	Ref	Ref
Other	17	1	5.6	1.19	0.14–9.96	1.90	0.19–18.79
Occupational status at amputation							
In paid employment ^f	124	8	6.1	1.00	Ref	NI	NI
Not in paid employment ^g	19	2	9.5	1.63	0.32–8.27	NI	NI
Child/youth/student ^h	56	0	0.0	0.20 ⁱ	0.00–1.35 ⁱ	NI	NI
Gross yearly income at amputation ^j							
No income	49	1	2.0	1.00	Ref	NI	NI
<300 000 NOK	99	5	4.8	2.48	0.28–21.77	NI	NI
≥300 000 NOK	33	2	5.7	2.97	0.26–34.10	NI	NI
Age at amputation							
<60 years	187	7	3.6	1.00	Ref	Ref	Ref
≥60 years	12	3	20.0	6.68	1.53–29.14	5.78	1.14–29.33
Level of amputation ^k							
Forearm/wrist	130	1	0.8	1.00	Ref	Ref	Ref
Elbow/upper arm/shoulder/forequarter	69	8	10.4	15.07	1.85–122.99	12.76	1.53–106.42
Amputation of dominant hand							
No	101	4	3.8	1.00	Ref	Ref	Ref
Yes	95	6	5.9	1.60	0.44–5.83	1.19	0.28–5.03
Cause of amputation ^l							
Traumatic	170	5	2.9	1.00	Ref	NI	NI
Other ^m	27	3	10.0	2.78	0.85–16.73	NI	NI

Abbreviations: cOR, Crude odds ratio; aOR, Adjusted odds ratio; CI, Confidence interval; Ref, Reference category; NI, Not included in adjusted model; NOK, Norwegian kroner.

^a $n = 5$ who were waiting for their first prosthesis at survey were excluded from these analyses (total unilateral $n = 214$).

^bCrude effect estimates: Logistic regression analysis, one independent variable in the model at a time.

^cAdjusted effect estimates: Logistic regression analysis, estimates adjusted for the included covariates, included $n = 205$.

^dPrimary prosthesis use, these amputees started to wear prostheses.

^ePrimary prosthesis rejection, these amputees have never worn prostheses.

^f $n = 109$ employed, $n = 23$ self-employed.

^g $n = 3$ work at home, $n = 7$ disability pension, $n = 7$ retired, $n = 2$ unemployed, $n = 2$ occupational rehabilitation.

^h $n = 26$ students, $n = 30$ child/youth.

ⁱExact values calculated with Stata.

^j $n = 189$ (crude estimate).

^k $n = 208$ (crude estimate).

^l $n = 205$ (crude estimate).

^mIncludes cancer, infection, arteriosclerosis, poor circulation, diabetes and overdose.

estimates did not change considerably by introducing it, it was kept in the model. Second, separate Cox regression analyses were conducted adding the covariate 'prosthetic type'. In these analyses, rejecters ($n = 10$) and current users (most worn prosthesis at survey, $n = 149$) of myoelectrical, body-powered and cosmetic prostheses were included. Rejecters were included if they had tried only one type of prosthesis before rejection. Calculating adjusted RR estimates only the variables 'gender', 'gross yearly income at amputation', 'age at amputation', 'level of amputation', 'amputation of dominant hand', 'prosthetic training' and 'prosthetic training enough' were included, since the inclusion of other variables did not lead to a valid model.

Results

Population characteristics

We identified 390 eligible amputees, of which 224 filled in questionnaires, yielding a 57.4% response rate. There were no

significant differences between the questionnaire responders and non-responders for the variables compared, implying that the questionnaire responders may be considered as representative of the sampling frame. Demographic factors and amputation-related factors for the questionnaire responders are shown in Table III.

Rates of prosthesis use and rejection

Two-hundred and nine (93.3%, 95% CI = 90.0–96.6) of the questionnaire responders stated that they had worn prostheses. Of these, 181 (86.6%, 95% CI = 82.0–91.2) were still prosthesis users at survey and 28 (13.4%, 95% CI = 8.8–18.0) had rejected their prostheses after some time of prosthesis use. 15 of the questionnaire responders had never worn prostheses, but only 10 (4.5% of 224, 95% CI = 1.8–7.2) were genuine primary prosthesis rejecters, detailed below. Details on prosthesis use, primary and secondary prosthesis rejection are given in Figure 1.

Table II. Secondary prosthesis rejection: effects of demographic, amputation-related and prosthesis-related factors in unilateral amputees ($n = 199$).

Independent variable	Secondary prosthesis rejection			Crude effect estimates ^a		Adjusted effect estimates ^b	
	Continued use (n) ^c	Rejection (n) ^d	% rejection	cRR	95% CI for cRR	aRR	95% CI for aRR
Gender							
Men	148	19	11.4	1.00	Ref	Ref	Ref
Women	24	8	25.0	2.39	1.04–5.47	4.58	1.21–17.39
Ethnicity (native country)							
Norway	156	26	14.3	1.00	Ref	Ref	Ref
Other	16	1	5.9	0.54	0.07–3.98	0.00	NA
Occupational status at amputation							
In paid employment ^e	109	15	12.1	1.00	Ref	Ref	Ref
Not in paid employment ^f	15	4	21.1	2.23	0.74–6.75	0.89	0.06–14.14
Child/youth/student ^g	48	8	14.3	0.97	0.41–2.30	1.82	0.22–15.07
Gross yearly income at amputation ^h							
No income	43	6	12.2	1.00	Ref	Ref	Ref
<300 000 NOK	87	12	12.1	1.24	0.46–3.31	2.41	0.25–23.21
≥300 000 NOK	27	6	18.2	2.80	0.87–9.01	5.39	0.32–90.84
Age at amputation							
<20 years	64	8	11.1	1.00	Ref	Ref	Ref
20–39 years	69	14	16.9	1.81	0.76–4.34	0.86	0.10–7.71
40–59 years	29	3	9.4	1.28	0.34–4.90	1.85	0.14–23.83
≥60 years	10	2	16.7	3.47	0.71–17.06	1.83	0.05–61.82
Level of amputation							
Forearm/wrist	122	8	6.2	1.00	Ref	Ref	Ref
Elbow/upper arm/shoulder/forequarter	50	19	27.5	5.54	2.42–12.68	5.71	1.70–19.16
Amputation of dominant hand ⁱ							
No	86	15	14.9	1.00	Ref	Ref	Ref
Yes	83	12	12.6	0.92	0.43–1.97	0.69	0.23–2.08
Cause of amputation ^j							
Traumatic	146	24	14.1	1.00	Ref	Ref	Ref
Other ^k	24	3	11.1	0.90	0.27–2.98	0.27	0.03–2.43
Time from amputation to first fitting ^l							
<3 months	45	5	10.0	1.00	Ref	Ref	Ref
3–5.9 months	44	9	17.0	1.69	0.57–5.03	2.22	0.47–10.49
6–8.9 months	25	5	16.7	1.62	0.47–5.59	1.50	0.27–8.28
9–11.9 months	22	3	12.0	1.05	0.25–4.40	1.44	0.12–16.70
≥12 months	28	2	6.7	0.64	0.12–3.28	0.63	0.04–10.63
Prosthetic training ^m							
Yes	111	16	12.6	1.00	Ref	Ref	Ref
No	51	9	15.0	1.34	0.59–3.04	1.42	0.38–5.25
Prosthetic training enough to meet needs ⁿ							
Yes	89	10	10.1	1.00	Ref	Ref	Ref
No	46	10	17.9	1.95	0.60–6.41	1.95	0.60–6.41

Abbreviations: cRR, Crude relative risk; aRR, Adjusted relative risk; CI, Confidence interval; Ref, Reference category; NA, Not applicable; NOK, Norwegian kroner.

^aCrude effect estimates: Cox regression analysis, one independent variable in the model at a time.

^bAdjusted effect estimates: Cox regression analysis, estimates adjusted for all the covariates, included $n = 138$.

^cThese amputees still used prostheses at survey.

^dSecondary prosthesis rejection, these amputees had rejected their prostheses after some time of use.

^e $n = 102$ employed, $n = 22$ self-employed.

^f $n = 3$ work at home, $n = 7$ disability pension, $n = 5$ retired, $n = 2$ unemployed, $n = 2$ occupational rehabilitation.

^g $n = 26$ students, $n = 30$ child/youth.

^h $n = 181$ (crude estimate).

ⁱ $n = 196$ (crude estimate).

^j $n = 197$ (crude estimate).

^kIncludes cancer, infection, arteriosclerosis, poor circulation, diabetes and overdose.

^l $n = 188$ (crude estimate).

^m $n = 187$ (crude estimate).

ⁿ $n = 155$ (crude estimate).

First referral and prosthetic training

First referral Only 74.8% of the 209 ULAs who had used prostheses after the amputation stated that they were referred to their first prosthetic fitting from the hospital where the

amputation took place. 8.7% were referred from their general practitioner and 4.9% stated that they contacted a prosthetic supplier themselves. In 4.4%, the first referral was conducted by a rehabilitation centre and in 2.4% by another hospital

than where the amputation took place. 1.0% was referred by a physiotherapist outside the hospital and 3.9% by 'others',

Table III. Questionnaire responders: demographic factors and amputation-related factors (n = 224^a).

Variable	
Gender (% men)	83.5
Age at survey, mean (years)	53.7
Ethnicity; native country (% Norwegian)	90.2
Place of residence, region (%)	
Eastern Norway	48.7
Western Norway	24.1
Northern Norway	10.7
Trøndelag (≈Mid Norway)	8.9
Southern Norway	7.6
Occupational status at amputation (%)	
In paid employment ^b	62.1
Not in paid employment ^c	11.1
Child/youth/student	26.8
Gross yearly income last year before amputation (%)	
No income	25.9
<300 000 NOK	55.7
≥300 000 NOK	18.4
Age at amputation, mean (years)	29.6
Time since amputation, mean (years)	24.0
Unilateral amputation, n (%)	214 (95.5)
Level of amputation (%) ^d	
Forearm/wrist	61.9
Elbow/upper arm/shoulder/forequarter	38.1
Unilateral, unknown level (n)	1
Bilateral, not same level on both sides (n) ^e	5
Amputation of dominant hand (% yes)	49.3
Cause of amputation (%)	
Trauma ^f	84.5
Other ^g	15.5

Abbreviation: NOK, Norwegian kroner.

^aThe number of responders was 201–224.

^bIncludes employed and self-employed.

^cIncludes work at home, disability pension, retired, unemployed and occupational rehabilitation.

^dPercentages include bilateral amputees with the same level of amputation on both sides, each counted as one amputee; % of 218.

^e2 wrist + upper arm, 1 wrist + elbow, 1 shoulder + elbow, 1 shoulder + lower arm.

^fIncludes occupational accidents, traffic accidents, war injury, explosion, high velocity, other accidents, torture and self-inflicted damage.

^gIncludes cancer, infection, arteriosclerosis/poor circulation/diabetes and overdose.

Table IV. Time from amputation to first prosthetic fitting (n = 196).

Period	Time from amputation to first fitting (months)				
	n	Mean	95% CI	Median	Range
1940–49	8	67.6	26.5–108.7	76.3	5.3–124.0
1950–59	16	16.1	2.8–29.3	12.0	0.8–108.0
1960–69	22	7.7	5.0–10.4	6.6	0.0 ^a –24.0
1970–79	39	7.0	4.8–9.2	5.0	0.0 ^a –42.0
1980–89	40	5.2	4.1–6.4	4.1	0.5–16.5
1990–99	40	20.4 ^b	9.4–31.3	6.8	0.8–135.0
2000–07	31	7.0	3.1–10.9	4.0	0.3–60.0

^aTime 0.0 indicates fitting within the first month post-amputation.

^bA very long time from amputation to first prosthetic fitting (135 months) in one amputee resulted in an unusually long mean first fitting time calculated for this ten-year period.

including their garrison doctor, their employer, their mother and a refugee organization.

Fitting time frame Among the primarily prosthesis-wearing ULAs in our material the mean time from the amputation to the first prosthetic fitting was 12.7 months (95% CI = 9.3–16.0), median 6.0 months, range 0.0–135.0 months. Analysing the time from the amputation until the first prosthetic fitting in ten-year periods from 1940 to 2007, we found that those fitted before 1960 had experienced long first fitting times whereas the median time from the amputation until the first prosthetic fitting has been relatively stable from 1960 until today. The range of the first fitting times was substantial in all ten-year periods. Mean times, median times and range for each ten-year period are shown in Table IV.

Prosthetic training Hundred and ninety six of the 209 amputees who had worn prostheses after the amputation gave information on prosthetic training. Of these, only 135 (68.9%, 95% CI = 62.4–75.4) stated that they had received prosthetic training and only 104 (62.7%, 95% CI = 55.3–70.0, valid n = 166) stated that they had received sufficient prosthetic training to meet their needs.

Primary prosthesis rejection

Descriptive analysis Fifteen amputees stated that they had never worn prostheses. Five of these amputees (33.3% or 2.2% of the questionnaire responders) stated that the reason for non-use was that they were recently amputated, waiting for their first prosthesis. Among the remaining 10 ULAs, the reasons given for rejection were that the prostheses available did not meet the amputee's needs (30.0%) or perceived lack of need (20.0%). One amputee (10.0%) had not used prostheses for medical reasons. The last four (40.0%) stated a non-specified 'other reason' for non-use.

Excluding the five amputees waiting for their first prosthesis, 10 amputees had genuinely primarily rejected prosthesis use. All primary prosthesis-rejecting amputees in our material were unilateral, yielding a primary rejection rate of 10/209 = 4.8% in unilateral amputees and of 0/10 = 0.0% in bilateral amputees. Using an exact method, we estimated an OR of 1.48 (95% CI = 0.00–10.40) for rejection in bilateral compared to unilateral ULAs, i.e. not a statistically significant difference as judged by the confidence interval.

Multivariate analysis (unilateral amputees) The effects of demographic and amputation-related factors on primary prosthesis rejection in unilateral major ULAs are shown in Table I. We found that ULAs amputated at or after the age of 60 years were 5.8 times more likely to reject prosthesis use than those amputated at a younger age, and that through or above-elbow amputees (proximal amputees) were 12.8 times more likely to reject prosthesis use than below-elbow amputees.

Secondary prosthesis rejection

Descriptive analysis In our material, 28 amputees started as prosthesis users but rejected their prosthesis at some time before the survey; 27 of these were unilateral amputees

(96.4%, 95% CI = 89.6–100.0) and one was a bilateral amputee (3.6%, 95% CI = 0.0–10.4), yielding a secondary prosthesis rejection rate of 27/199 = 13.6% (95% CI = 8.8–18.3) among unilateral amputees and of 1/10 = 10.0% (95% CI = 0.0–28.6) among bilateral amputees. Using an exact method, we estimated an OR for rejection in bilateral vs unilateral amputees of 0.71 (95% CI = 0.02–5.49), i.e. not a statistically significant difference as judged by the confidence interval.

Among the unilateral amputees, the mean time from first prosthetic fitting to rejection was 6.7 years (95% CI = 3.2–10.2), range 0.0–32.0 years; time 0.0 indicating rejection within the first month after first fitting. Only 10 unilateral amputees had tried only one type of prosthesis before rejection (three myoelectric, six body-powered and one cosmetic). Among the rest of the rejecters seven had tried two different prosthetic types and three had tried more than two different types whereas seven did not give information on the type of prosthesis worn before rejection.

The main reasons for secondary prosthesis rejection stated by the unilateral amputees were dissatisfaction with prosthetic comfort, function or control, listed as sole or partial reason for rejection by 13 (comfort, function) and six (control) amputees, respectively. Five amputees stated lack of prosthetic training as a partial reason for rejection. Three amputees had ceased prosthesis use solely due to pain and pain was also a partial reason for rejection in another amputee. Two amputees stated that they rejected their prosthesis because 'the prosthesis was in the way', one that stated that he did not need the prosthesis because he managed well using his contralateral arm and one stated that the reason for rejection was that 'the shoulder is too weak'. Eleven amputees stated more than one reason for rejection, nine of these stated three or more reasons.

Only one bilateral amputee in our sample had secondarily rejected prosthesis use. He was a male, traumatic amputee, in paid employment at amputation and with amputation level through the wrist and through the forearm. He had tried body-powered prostheses with hand and hook bilaterally before rejection. The time from the first prosthetic fitting until rejection is not known. He stated that he ceased prosthesis use due to a combination of dissatisfaction with prosthetic comfort, function and control, specified by the comment 'I got tired in the shoulders due to wearing the prostheses'.

Multivariate analysis without prosthetic type (unilateral amputees) The effects of demographic, amputation-related and prosthesis-related factors on secondary prosthesis rejection in unilateral major ULAs are shown in Table II. We found that secondary prosthesis rejection was more likely in women and in amputees with proximal amputations. The relative risk of secondary prosthesis rejection in proximal amputees vs below-elbow amputees was 5.7 and the relative risk of secondary prosthesis rejection in women vs men was 4.6.

Multivariate analysis including prosthetic type (unilateral amputees) The effect of amputation level on secondary prosthesis rejection was also significant in the model

controlling for prosthetic type (n included in the adjusted model = 115): adjusted RR for rejection in through or above-elbow vs below-elbow amputees was 16.26 (95% CI = 1.42–186.61). In this model, there were no significant effects of any of the other covariates, although a tendency to higher rejection rates in users of body-powered than in users of cosmetic prostheses was observed (adjusted RR = 17.63, 95% CI = 0.92–339.72).

Discussion

Main findings

Primary prosthesis rejection was found in 4.5% of acquired major ULAs whereas 13.4% of acquired major ULAs primarily wearing prostheses had discontinued prosthesis use. There was no clear evidence of a difference in the rates of prosthesis rejection between unilateral and bilateral amputees.

The main reasons reported for primary non-wear were a perceived lack of need and discrepancies between the perceived need and the prostheses available. The main reasons reported for secondary prosthesis rejection were dissatisfaction with prosthetic comfort, function and control.

In unilateral ULAs, primary prosthesis rejection was more likely among those amputated at high age and among those with proximal amputations, whereas secondary prosthesis rejection was more likely in proximal ULAs and in women. The effect of amputation level on secondary prosthesis rejection was significant also in a model controlling for prosthetic type. A tendency to higher rejection rates in wearers of body-powered compared to cosmetic prostheses was observed, though there was no clear evidence of such a difference.

Rejection rates

We found that most acquired major ULAs had worn prostheses for some time after the amputation, consistent with the review by Jones and Davidson, stating that 'almost all patients wish to be fitted with a prosthesis (...), even if in the long term they do not remain a prosthetic user' [15]. The estimated prevalence of primary rejection of 4.5% in our study is consistent with the findings of Datta and Ibbotson who found that 96% of the ULAs (acquired and congenital) were provided with prostheses [16], but lower than reported by Biddiss and Chau, who found a primary prosthesis rejection rate of 14% in an internet survey of acquired and congenital limb loss [17]. The higher prevalence of primary prosthesis rejection noted by Biddiss and Chau may be attributable to the inclusion of partial hand and wrist amputees in their study, this group being overrepresented among those who had never worn a prosthesis [17]. Berger demonstrated rates of primary prosthesis rejection of 16–30% according to the amputation level [14], but these data were published in 1958 and are hardly comparable with today's prosthetic practice.

The 181 amputees wearing prostheses at survey constituted 80.8% of the questionnaire responders, a prosthetic usage rate within that earlier found in Northern hemisphere studies, though earlier findings vary considerably [7,8]. In accordance

with the recent findings of Biddiss and Chau [17], the rejection rates for unilateral and bilateral amputees did not differ significantly in our sample.

Most frequent reasons for prosthesis rejection

From earlier literature, it is clear that prosthesis acceptance and rejection is complex and dependent on a number of personal, contextual and technological variables and that the prosthetic outcome for ULAs vary considerably [9,15]. Some studies have reported the majority of prosthesis rejection due to external factors such as delayed prosthetic delivery, medical complications or lack of need; others have reported 91% of myoelectric rejections resulting from prosthesis-related problems involving comfort, lack of function and durability [7], suggesting that the decision to accept or reject a prosthesis may be extremely individualised [18].

In our study, the main reasons reported for primary prosthesis non-wear were a perceived lack of need and discrepancies between perceived need and the prostheses available, illustrating the importance of matching the prosthetic devices to the needs of the individual to ensure successful prosthetic rehabilitation [9,17,19–22]. Also, in our experience, promoting realistic expectations of prosthetic performance early in the rehabilitation process is of great importance to prosthesis acceptance. This is emphasised by Biddiss and Chau's finding that involving the amputee in the selection of his/her prosthesis increases the likelihood of acceptance by nearly eight times [18]. Currently, the rehabilitation of ULAs in Norway is not standardised. Its organisation differs between the sites offering specialised ULA rehabilitation and amputees are not routinely referred to rehabilitation or prosthetic fitting at discharge from the surgical ward [6]. In our material, the mean time from the amputation until the first prosthetic fitting was long and only about 75% had been referred to their first prosthetic fitting from the hospital where the amputation took place. To ensure optimal prosthetic rehabilitation including early adequate information and assessment of the amputee's needs, such referral should probably be mandatory.

The main reasons for secondary prosthesis rejection stated by the amputees in our sample – dissatisfaction with prosthetic comfort, function and control – are in accordance with the recent findings of Biddiss and Chau [17] and McFarland et al. [23] as well as with a number of earlier studies [8,22,24–26]. These findings suggest that further development of prosthesis quality (improving comfort and function) as well as individualised prosthetic training (improving function and control) may increase long-term prosthesis use. Existing literature is however ambiguous as to the importance of each factor, and different reasons for rejection have been reported in cosmetic, body-powered and electric devices [7]. Also, the rejection rates for passive, myoelectric and body-powered prostheses vary considerably in the literature and it is difficult to make comparisons between the wear patterns of passive and active devices, as one type often supplements the other, and the use of multiple

prostheses is widely recommended [7]. As discussed below, controlling for prosthetic type in the multivariate analyses of rejection was also a challenge in the present study. The pattern of and reasons for rejection in each prosthetic type could probably be better explored in a prospective study including amputees at first prosthetic fitting.

Background factors associated with prosthesis rejection

The background factors associated with prosthesis use and rejection are not clear, and even less is known on their relative importance in the decision to accept or reject prosthesis use [7,18]. In the earlier literature, we could only find one multivariate analysis of factors associated with upper-limb prosthesis rejection [18]. Thus, conducting multivariate analyses in a population-based sample, our study provides an important contribution to the existing body of knowledge on prosthesis use and rejection in ULAs.

Furthermore, most previous studies have looked solely at secondary prosthesis rejection or have not separated primary and secondary rejection [2,5,7,8,21–25,27–37]. Our findings indicate that somewhat different factors are of importance to primary and secondary prosthesis rejection, and thus that such a segregation of rejection may be clinically important in ULA rehabilitation.

Our findings that primary prosthesis rejection was more frequent in ULAs amputated at high age are consistent with the recent findings of Biddiss and Chau [17]. These also found that limb loss distal to the wrist was more frequent among the primary rejecters [17]. This amputation level was not included in our material. However, we found that primary prosthesis rejection was more frequent in through or above-elbow amputees. Combining these findings it may be deduced that upper limb loss at the transradial level seems to increase the chance of primary prosthesis use.

Our findings of a significant effect of proximal amputation level on secondary prosthesis rejection is strongly supported in the literature [8,9,14,17,18,24,25,30,38,39]. Acceptance of a prosthesis generally decreases at higher levels of limb loss where the devices are heavier and require greater energy expenditure to operate [9]. Interestingly, we found that the effect of a proximal amputation level was strong also controlling for prosthetic type, i.e. regardless of whether the amputee had been fitted with a cosmetic, body-powered or myoelectric prosthesis before rejection.

The majority of the earlier literature reports no correlation between gender and prosthesis rejection [9], though in a recent study by Biddiss and Chau assessing critical factors in device abandonment [17], it was found that women were more likely to reject prostheses than men. In our study, we found that women were more likely to reject their prosthesis than men, controlling for several demographic, amputation-related and prosthesis-related factors. That the effect of gender was non-significant controlling for prosthetic type may be attributed to lower statistical power for the findings in this sample, as discussed below.

The lack of significant effects on secondary prosthesis rejection of other factors analysed in our study is mainly in

accordance with earlier literature [9]. However, in the recent work of Biddiss and Chau fitting time frame emerged as the predominant predictor of prosthesis use; adult acquired ULAs fitted within six months of amputation were found to be 16 times more likely to continue prosthesis use [18]. The concept of a 'golden period of fitting' of less than 30 days post amputation was introduced by Malone [40] and the effect of early prosthetic fitting on prosthesis use and satisfaction has also been shown by others [27,41], although in sum, the effect of time lapse between amputation and first fitting remains unsolved [9].

Earlier literature suggests that the effect of prosthetic training on prosthesis use and rejection is uncertain [9], although a desire for more customised and extensive training adjusted to meet individual needs is evident [5,37]. As much as 44% of non-wearers have been reported to be dissatisfied with the preparation and training received [9], in accordance with our findings that only about 60% reported to have received prosthetic training enough to meet their needs. Insufficient follow-up and lack of ongoing training have been implicated as possible causes for high rejection rates [9]. However, training protocols vary greatly between rehabilitation sites, and though several studies have noted improved functional use as a result of training programs undergone, greater proficiency did not necessarily equate to increased wearing time or acceptance [5,9,32,42,43]. Clearly, this is an important area of further research.

Strengths and limitations

A relatively low response rate may have led to selection bias in our sample. However, our comparison of questionnaire responders and non-responders indicates that the responders may be considered as representative of the sampling frame.

Studying prosthesis use and rejection, we used the amputees' own definition of use, regardless of the frequency of use or the type of prosthesis. Some very occasional users would have been excluded with more strict criteria of use, indicating that our estimates of rejection rates may be conservative. This may also be the result of including amputees even after a very short period of follow-up; it has been shown that a short period of follow-up may give false positive outcome in terms of prosthesis use [24]. However, for most amputees in our sample, first prosthetic fitting and eventual rejection took place several years before the survey. A few amputees who were waiting for their first prosthesis at survey were excluded from our estimate of primary prosthesis rejection.

Our findings regarding the effect of prosthetic type must be interpreted with caution. Due to insufficient data regarding the wearing times and sequence of different types of prostheses worn before rejection, only rejecters that had worn only one type of prosthesis were included in these analyses. Also, the comparison with the most worn prosthesis among current prosthesis users did not take into consideration that these users may also be wearing other types of prostheses

or that they may have rejected other prosthetic types before the survey.

Information bias may have been introduced by missing data, especially regarding time until the first prosthetic fitting. Our imputation procedures may not fully have compensated for such a bias. Also, a relatively high proportion of missing data regarding some of the other variables (i.e. prosthetic training and prosthetic type(s) used before rejection) may have led to reduced statistical power in the multivariate analyses. Moderate effects on prosthesis rejection may thus have been left undetected.

It may be considered a weakness of our study that we have included amputees who were amputated during a broad time frame, as both prostheses and rehabilitation routines have changed considerably over the last 70 years. However, this may also be considered a strength of the study, contributing to a relatively comprehensive overview of the prosthesis use and experiences of the population of adult major ULAs resident in Norway today.

Strengths of the study include a population-based questionnaire sample including amputees from the entire country, the collection of data regarding a broad range of the factors that may influence prosthesis use and rejection, the inclusion of users of different prosthetic types and conducting formal statistical analyses including multivariate analyses.

Conclusions

Primary prosthesis rejection was found in 4.5% whereas 13.4% of primary prosthesis-wearing ULAs had discontinued prosthesis use. There was no clear evidence of a difference in the rates of prosthesis rejection between unilateral and bilateral amputees.

In unilateral ULAs, we found an increased risk of primary prosthesis rejection in older adults and in through or above-elbow amputees, suggesting that referral to specialised rehabilitation may be especially important in these ULAs. Furthermore, our findings suggest that emphasising individual needs will facilitate successful prosthetic fitting. To ensure optimal individualised prosthetic rehabilitation including early adequate information and assessment of the amputee's needs, ULA rehabilitation in Norway should be standardised, and referral to prosthetic fitting should probably be mandatory at or before discharge from the hospital where the amputation took place.

In unilateral ULAs, secondary prosthesis rejection was more likely in through or above-elbow amputees and in women, guiding clinicians to be especially aware of these ULAs in the planning of long-term ULA prosthetic rehabilitation. Furthermore, our findings suggest that further development of prosthesis quality as well as individualised prosthetic training may increase long-term prosthesis use. Further studies of the effect of prosthetic training and of the reasons for rejection of different prosthetic types are suggested.

Declaration of Interest: The authors report no conflicts of interest. Our work was funded in part by Innlandet Hospital

Trust and in part by the Norwegian Extra Foundation for Health and Rehabilitation.

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